

Equation with $u(n)$;

$$T(n)u(n) = 2T(n-1)u(n-1) + n2^n u(n) + \delta(n)$$

$$\mathcal{Z}\{T(n-1)u(n-1)\} = z^{-1}X(z)$$

$$\mathcal{Z}\{n2^n u(n)\} = \frac{2z^{-1}}{(1-2z^{-1})^2}$$

$$\mathcal{Z}\{\delta(n)\} = 1;$$

$$X(z) = 2(z^{-1}X(z)) + \frac{2z^{-1}}{(1-2z^{-1})^2} + 1$$

$$X(z)[1-2z^{-1}] = 1 + \frac{2z^{-1}}{(1-2z^{-1})^2}$$

$$X(z)[1-2z^{-1}]^2 = \frac{1-2z^{-1}+4z^{-2}}{(1-2z^{-1})^2}$$

$$X(z) = \frac{1-2z^{-1}+4z^{-2}}{(1-2z^{-1})^3}$$

$$X(z) = \frac{1}{(1-2z^{-1})^3} - \frac{(2z^{-1}+4z^{-2})}{(1-2z^{-1})^3}$$

$$\frac{1}{(1-2z^{-1})^3} - \frac{(2z^{-1}(1-2z^{-1}))}{(1-2z^{-1})^3 \cdot 2}$$

$$\frac{1}{(1-2z^{-1})^3} - \frac{-1(1-2z^{-1}+1)}{(1-2z^{-1})}$$

$$X(z) = \frac{1}{(1-2z^{-1})^3} - \frac{1}{(1-2z^{-1})^2} + \frac{1}{(1-2z^{-1})} \quad \left| \begin{array}{l} \text{denominator} \\ 1-2z^{-1}=0 \end{array} \right.$$

Inverse transform;

Region of convergence = $z=2$

$$\mathcal{Z}^{-1} \left\{ \frac{1}{(1-az^{-1})^k} \right\} = \left(\frac{(n+k-1)!}{(k-1)!n!} \right) a^n u(n)$$

$$\Rightarrow T(n) = \left[\frac{(n+1)(n+2)}{2} - (n+1) + 1 \right] 2^n u(n)$$

at $k=3$; $a=2$

$$\mathcal{Z}^{-1} \left\{ \frac{1}{(1-2z^{-1})^3} \right\} \Rightarrow \frac{(n+2)!}{2!n!} \Rightarrow \frac{(n+2)(n+1)}{2} 2^n$$

$$\mathcal{Z}^{-1} \left\{ \frac{1}{(1-2z^{-1})^2} \right\} \Rightarrow \frac{(n+1)!}{n!} \Rightarrow (n+1) \cdot 2^n$$

$$\mathcal{Z}^{-1} \left\{ \frac{1}{(1-2z^{-1})} \right\} \Rightarrow \frac{n!}{n!} = 1 \cdot 2^n$$

$$T(n) = \left[\frac{(n+2)(n+1)}{2} - (n+1) + 1 \right] 2^n$$

$$T(n) = \frac{(n+1)n+2}{2} 2^n$$

$$T(n) = \frac{n(n+1)}{2} \cdot 2^n + 1 \rightarrow \text{order} = n^2 2^n$$